

The exact Riesz bounds of the ReLU tent-function basis

A full solution of the optimal-bound problem in arXiv:2303.00076

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Lane 14, agent `agent_lane_14`

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Status: claimed full solution; likely valid, pending human review. The optimal uniform Gram spectral interval is $[2/3, 3/2]$. Thus the published lower estimate $1/2$ improves to the sharp value $2/3$, while the upper estimate $3/2$ is already sharp. The same exact constants hold for the normalized multivariate system in every dimension.

1 Source and open problem

The source is Cornelia Schneider and Jan Vybiral, *A multivariate Riesz basis of ReLU neural networks*, arXiv:2303.00076, published in *Applied and Computational Harmonic Analysis* **68** (2024), article 101605, DOI [10.1016/j.acha.2023.101605](https://doi.org/10.1016/j.acha.2023.101605) [1]. The question occurs in Remark 2.3(3) on PDF page 5, immediately after Theorem 2.2.

Writing

$$\overline{\mathcal{R}}_1^N = \{1\} \cup \{\sqrt{3}\mathcal{C}_k, \sqrt{3}\mathcal{S}_k : 1 \leq k \leq N\},$$

the source proves that its Gram spectrum lies in $[1/2, 3/2]$, uniformly in N , and asks whether those bounds are optimal. It conjectures that there is room for improvement [1, Remark 2.3(3)].

2 Proof intuition

The Gram matrix is not merely sparse; it is multiplicative. Indices having different powers of two are orthogonal. Within one fixed 2-adic block, remove the common power of two and write the remaining odd integers by their prime-exponent vectors. The normalized cosine kernel then factors prime by prime:

$$\frac{\gcd(m, n)^4}{m^2 n^2} = \prod_{p \text{ odd}} p^{-2|v_p(m) - v_p(n)|}.$$

For one prime this is the Toeplitz kernel $r^{|a-b|}$, whose Fourier symbol has exact range $[(1-r)/(1+r), (1+r)/(1-r)]$. Tensoring over the odd primes and evaluating the Euler products gives $[2/3, 3/2]$.

The inequalities alone would not prove optimality. To reach both endpoints, take finite rectangular grids of prime exponents. Constant coefficients concentrate the local Fourier mass at angle 0; alternating coefficients concentrate it at angle π . These are genuine finite coefficient vectors, hence occur inside sufficiently large initial Gram matrices. The sine block is a diagonal sign conjugate of the cosine block. Finally, in several variables, distinct primitive lattice rays are orthogonal and every ray carries exactly the same univariate block.

Next, we combine Lemma 2.1 with the Gershgorin circle theorem and provide an alternative proof of [7] Theorem 6.2], which we restate as follows.

Theorem 2.2. *The system $\mathcal{R}_1 := \{1\} \cup \{C_k, S_k : k \in \mathbb{N}\}$ is a Riesz basis of $L_2([0, 1])$.*

Before we come to the proof, several remarks seem to be in order.

- Remark 2.3.**
1. We will actually show, that the Riesz constants of the L_2 -normalized system $\overline{\mathcal{R}}_1 := \{1\} \cup \{\sqrt{3}C_k, \sqrt{3}S_k : k \in \mathbb{N}\}$ can be chosen as $A = 1/2$ and $B = 3/2$.
 2. We divide the proof of Theorem 2.2 into several steps. In the first two steps, we show that the truncated system $\overline{\mathcal{R}}_1^N := \{1\} \cup \{\sqrt{3}C_k, \sqrt{3}S_k : k \leq N\}$ forms a Riesz sequence (i.e., that it satisfies (1)) with $A = 1/2$ and $B = 3/2$ being independent of $N \in \mathbb{N}$. The first step reduces this question to spectral properties of the Gram matrix of $\overline{\mathcal{R}}_1^N$ and in the second step we apply the Gershgorin theorem to bound this spectrum. The third step describes how we pass to the limit $N \rightarrow \infty$ to deduce that $\overline{\mathcal{R}}_1$ is also a Riesz sequence. Finally, the fourth step shows that $\overline{\mathcal{R}}_1$ is also a basis, i.e., that its closed linear span is $L_2([0, 1])$.
 3. Our proof shows that the spectrum of the Gram matrix of $\overline{\mathcal{R}}_1^N$ (for arbitrary $N \in \mathbb{N}$) is contained in $[1/2, 3/2]$. We leave it as an open problem to find out if these bounds are actually optimal. Supported by numerical evidence (see Figure 3 for details), our conjecture is that there is indeed some space for improvement.

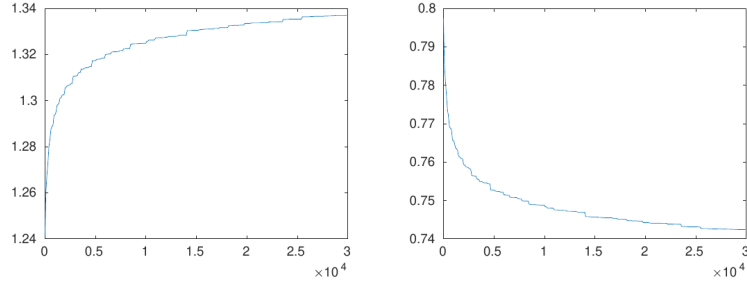


Figure 3: The largest (left) and the smallest (right) eigenvalue of $(\langle C_i, C_j \rangle)_{i,j=1}^N$ for $1 \leq N \leq 3 \cdot 10^4$.

Figure 1: Full-width crop of Theorem 2.2 and Remark 2.3 from source PDF page 5. The source's plots show the *unnormalized* cosine Gram eigenvalues; see Corollary 3.

3 Statement of the result

Let G_N be the Gram matrix of the ordered normalized system $\overline{\mathcal{R}}_1^N$. Thus its diagonal entries are one. The source uses Riesz constants in the squared-norm convention

$$A \sum_j |a_j|^2 \leq \left\| \sum_j a_j u_j \right\|_2^2 \leq B \sum_j |a_j|^2.$$

Theorem 1 (Exact uniform constants). *For every $N \geq 1$,*

$$\sigma(G_N) \subseteq [2/3, 3/2].$$

Moreover,

$$\inf_{N \geq 1} \lambda_{\min}(G_N) = \frac{2}{3}, \quad \sup_{N \geq 1} \lambda_{\max}(G_N) = \frac{3}{2}.$$

Consequently the optimal Riesz bounds of the infinite normalized univariate system are exactly $A = 2/3$ and $B = 3/2$.

For every $d \geq 1$, the normalized multivariate system of [1],

$$\{1\} \cup \{\sqrt{3}C(\alpha \cdot x) : \alpha \in \mathbb{Z}^d, \alpha \triangleright 0\} \cup \{\sqrt{3}S(\alpha \cdot x) : \alpha \in \mathbb{Z}^d, \alpha \triangleright 0\},$$

has the same exact optimal Riesz bounds $2/3$ and $3/2$, independently of d . Here $\alpha \triangleright 0$ means that the first nonzero coordinate of α is positive.

4 Proof

4.1 The source Gram formulas

Put $C_N = (3\langle \mathcal{C}_i, \mathcal{C}_j \rangle)_{1 \leq i, j \leq N}$ and define S_N analogously. Lemma 2.1 of [1] gives

$$G_N = [1] \oplus C_N \oplus S_N,$$

and states that the cosine entry is zero when $\nu_2(i) \neq \nu_2(j)$. When $\nu_2(i) = \nu_2(j) = a$, write $i = 2^a m$ and $j = 2^a n$ with m, n odd. Then

$$(C_N)_{ij} = \frac{\gcd(m, n)^4}{m^2 n^2} = \prod_{p \text{ odd}} p^{-2|v_p(m) - v_p(n)|}. \quad (1)$$

Only finitely many factors in the last product differ from one.

4.2 The exact one-prime inequality

Lemma 2. *Let $0 < r < 1$, and let $A_r = (r^{|u-v|})_{u, v \geq 0}$ act through its quadratic form on finitely supported vectors. Then*

$$\frac{1-r}{1+r} \|x\|_2^2 \leq \sum_{u, v \geq 0} r^{|u-v|} x_u \overline{x_v} \leq \frac{1+r}{1-r} \|x\|_2^2. \quad (2)$$

Both constants are optimal.

Proof. The Poisson-type symbol with Fourier coefficients $r^{|h|}$ is

$$f_r(\theta) = \sum_{h \in \mathbb{Z}} r^{|h|} e^{ih\theta} = \frac{1-r^2}{1-2r \cos \theta + r^2}.$$

For finitely supported x , Parseval gives

$$\sum_{u, v \geq 0} r^{|u-v|} x_u \overline{x_v} = \int_0^{2\pi} f_r(\theta) \left| \sum_{u \geq 0} x_u e^{iu\theta} \right|^2 \frac{d\theta}{2\pi}.$$

The minimum and maximum of f_r occur at π and 0 , respectively, and are precisely the two constants in (2).

For sharpness, take the normalized vectors $x_u = M^{-1/2}$ and $x_u = M^{-1/2}(-1)^u$ for $0 \leq u < M$. Their Rayleigh quotients are

$$q_+(r, M) = 1 + \frac{2}{M} \sum_{h=1}^{M-1} (M-h)r^h, \quad q_-(r, M) = 1 + \frac{2}{M} \sum_{h=1}^{M-1} (M-h)(-r)^h. \quad (3)$$

As $M \rightarrow \infty$, these tend to $(1+r)/(1-r)$ and $(1-r)/(1+r)$, respectively. $\square$

4.3 Tensoring over odd primes

Let F be a finite set of odd primes. On $\ell_2(\mathbb{N}_0^F) = \bigotimes_{p \in F} \ell_2(\mathbb{N}_0)$, define

$$A_F = \bigotimes_{p \in F} A_{p^{-2}}.$$

Repeated application of Lemma 2 gives

$$L_F I \leq A_F \leq U_F I, \quad L_F = \prod_{p \in F} \frac{1 - p^{-2}}{1 + p^{-2}}, \quad U_F = L_F^{-1}. \quad (4)$$

Now let x be any finitely supported coefficient vector on the odd positive integers. Choose F containing every prime divisor of every index in the support. The map $n \mapsto (v_p(n))_{p \in F}$ identifies its quadratic form from (1) with the restriction of the quadratic form of A_F . Therefore (4) applies even though the support need not be a rectangular exponent grid.

The Euler products are

$$\begin{aligned} \prod_{p \text{ odd}} (1 - p^{-2}) &= \frac{1/\zeta(2)}{1 - 2^{-2}} = \frac{8}{\pi^2}, \\ \prod_{p \text{ odd}} (1 + p^{-2}) &= \frac{\prod_{p \text{ odd}} (1 - p^{-4})}{\prod_{p \text{ odd}} (1 - p^{-2})} = \frac{12}{\pi^2}. \end{aligned}$$

Consequently

$$\prod_{p \text{ odd}} \frac{1 - p^{-2}}{1 + p^{-2}} = \frac{2}{3}, \quad \prod_{p \text{ odd}} \frac{1 + p^{-2}}{1 - p^{-2}} = \frac{3}{2}. \quad (5)$$

Since $L_F \geq 2/3$ and $U_F \leq 3/2$, every odd block is bounded between $2/3$ and $3/2$. Every fixed 2-adic block is a copy of the odd block, so

$$\frac{2}{3} I \leq C_N \leq \frac{3}{2} I \quad (6)$$

for every N .

4.4 The sine block is a sign conjugate

For $k \geq 1$, let $k = 2^{\nu_2(k)} k_{\text{odd}}$ and set

$$\varepsilon_k = \chi_4(k_{\text{odd}}), \quad \chi_4(t) = \begin{cases} 1, & t \equiv 1 \pmod{4}, \\ -1, & t \equiv 3 \pmod{4}. \end{cases}$$

Let $D_N = \text{diag}(\varepsilon_1, \dots, \varepsilon_N)$. We claim that

$$S_N = D_N C_N D_N. \quad (7)$$

Indeed, entries from different 2-adic blocks vanish on both sides. In one block write $i = 2^a m$, $j = 2^a n$, $d = \gcd(m, n)$, and $m_0 = m/d$, $n_0 = n/d$. The sign rule in Lemma 2.1 of [1] says that the sine entry is negative exactly when $(m_0 + n_0)/2$ is even. For coprime odd m_0, n_0 , this is equivalent to $\chi_4(m_0)\chi_4(n_0) = -1$. Since $\chi_4(d)^2 = 1$, this sign is $\varepsilon_i \varepsilon_j$, proving (7). Thus S_N and C_N have identical spectra. This similarity was also proved independently in the later paper [2]; here the displayed argument derives it directly from the source formula.

Combining (6), (7), and the scalar constant block proves $\sigma(G_N) \subseteq [2/3, 3/2]$.

4.5 Finite witnesses prove sharpness

Fix a finite set F of odd primes and $M \geq 1$. Consider the finite index set

$$E_{F,M} = \left\{ \prod_{p \in F} p^{a_p} : 0 \leq a_p < M \right\}.$$

Give these indices either constant coefficients or coefficients $(-1)^{\sum_{p \in F} a_p}$, and normalize by $|E_{F,M}|^{-1/2}$. The product structure makes their cosine Rayleigh quotients exactly

$$Q_+(F, M) = \prod_{p \in F} q_+(p^{-2}, M), \quad Q_-(F, M) = \prod_{p \in F} q_-(p^{-2}, M). \quad (8)$$

For fixed F , equation (3) yields

$$Q_+(F, M) \longrightarrow U_F, \quad Q_-(F, M) \longrightarrow L_F \quad (M \rightarrow \infty).$$

Then let F increase through the odd primes and use (5). For every $\delta > 0$, this produces a finite coefficient vector with quotient greater than $3/2 - \delta$, and another with quotient less than $2/3 + \delta$.

Every set $E_{F,M}$ lies inside $\{1, \dots, N\}$ for some finite N . Extend the witness by zeros on the other cosine indices and on the constant and sine blocks. The min–max principle therefore gives

$$\sup_N \lambda_{\max}(G_N) \geq \frac{3}{2}, \quad \inf_N \lambda_{\min}(G_N) \leq \frac{2}{3}.$$

Together with the already proved inequalities, both are equalities. This completes the univariate proof.

4.6 Multivariate upgrade

Every oriented nonzero lattice vector has a unique representation $\alpha = k\gamma$, where $k \in \mathbb{N}$ and $\gamma \in \mathbb{Z}^d$ is primitive and oriented. Lemma 3.1 of [1] states that functions indexed by non-collinear vectors are orthogonal, that cosine–sine cross terms vanish, and that along one ray $\{k\gamma : k \geq 1\}$ the normalized cosine and sine inner products are exactly the univariate ones. Hence, after reordering, every finite multivariate Gram matrix is a compression of the orthogonal direct sum

$$[1] \oplus \bigoplus_{\gamma \text{ primitive, oriented}} (C \oplus S).$$

The bounds $[2/3, 3/2]$ therefore hold in every dimension. The source already proves completeness, so these are Riesz-basis bounds. Sharpness is realized asymptotically on the single primitive ray generated by $(1, 0, \dots, 0)$, using the finite witnesses above. Theorem 1 is proved. $\square$

Corollary 3 (Normalization of the source plots). *For the unnormalized cosine and sine systems, whose Gram blocks are $(\langle \mathcal{C}_i, \mathcal{C}_j \rangle)$ and $(\langle \mathcal{S}_i, \mathcal{S}_j \rangle)$, the exact optimal uniform spectral interval is $[2/9, 1/2]$.*

Proof. The normalized blocks are three times the unnormalized blocks. $\square$

5 Verification

The proof is analytic. The included script `code/verify_bounds.py` performs three independent numerical audits:

1. it constructs normalized cosine Gram matrices for $N = 16, 32, 64, 128, 256, 512$ and checks all eigenvalues against $[2/3, 3/2]$;
2. it constructs $S_N = D_N C_N D_N$ and compares the complete cosine and sine spectra (worst mismatch: 0.000e+00 in the run);
3. on the first eight odd primes it evaluates the length-80 finite tensor witnesses, obtaining 0.681311350825 below and 1.467744146372 above, moving toward $2/3$ and $3/2$, respectively.

The finite spectra at $N = 512$ were $\lambda_{\min} = 0.773698855226$ and $\lambda_{\max} = 1.280150591375$. The reproducible command is

```
conda run --no-capture-output -n sandbox python \
    code/verify_bounds.py
```

These computations are consistency checks, not ingredients of the proof.

6 Novelty check, limitations, and review focus

Bounded novelty check. A bounded search through 17 August 2026 used the exact source title and authors, citation links, arXiv:2303.00076, “optimal Riesz bounds,” “Gram spectrum,” and the candidate constants $2/3$ and $3/2$. The strongest relevant follow-up found was Abdullo Kulbatov, Jan Lang, Cornelia Schneider, and Jan Vybíral, *Bases of Lebesgue spaces formed by neural networks*, arXiv:2511.23179 [2]. It proves the finite cosine/sine spectral coincidence and improves the known lower estimate to

$$\left\{ \frac{4\sqrt{6}}{\pi^2} \left(2 - \frac{\pi^2}{8} \right) \right\}^2 = 0.5787 \dots,$$

in both the univariate and multivariate settings, while retaining upper bound $3/2$. It does not derive $2/3$ or prove optimality. A later paper using the basis for Sobolev and Barron approximation likewise did not provide the exact constants in the checked material. No matching exact answer was found. This is a bounded negative search, not a claim of exhaustive novelty.

Limitations. The theorem determines the best constants uniform in N ; it does not give a closed formula for the extreme eigenvalues at each fixed N , nor rates at which selected initial sections approach the endpoints. The multivariate statement improves the constants for the precise lattice-ridge basis of [1]; it is not a claim about arbitrary ReLU dictionaries.

Human-review recommendation. Review the identification of a finitely supported odd-index quadratic form with a compression of the finite-prime tensor product, the order of the two limits in the explicit sharpness witnesses, and the primitive-ray direct-sum decomposition in the multivariate argument. Also verify that the source’s Riesz constants use the squared-norm convention, as adopted here.

References

- [1] C. Schneider and J. Vybíral, *A multivariate Riesz basis of ReLU neural networks*, Applied and Computational Harmonic Analysis **68** (2024), 101605; arXiv:2303.00076.
- [2] A. Kulbatov, J. Lang, C. Schneider, and J. Vybíral, *Bases of Lebesgue spaces formed by neural networks*, arXiv:2511.23179 (2025).